

PATHWAY PIRATES

[PROJECT GROUP - 5]

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PROJECT TITLE :-

**Hypoxia driven Translational
EGFR Upregulation vs Gene
Amplification: Do Divergent ERK
Kinetics Alter Inhibitor
Sensitivity?**

Introduction

Cancer develops when cells lose their normal ability to control growth and begin dividing uncontrollably. This process is regulated by several signaling pathways, and one of the most important is the Epidermal Growth Factor Receptor (EGFR) signaling pathway. When EGFR is activated, it triggers a series of proteins, including RAS, RAF, MEK, and ERK, which help cells grow, divide, and survive. In many types of cancer, EGFR is overexpressed, causing these signals to remain active and leading to uncontrolled tumor growth.

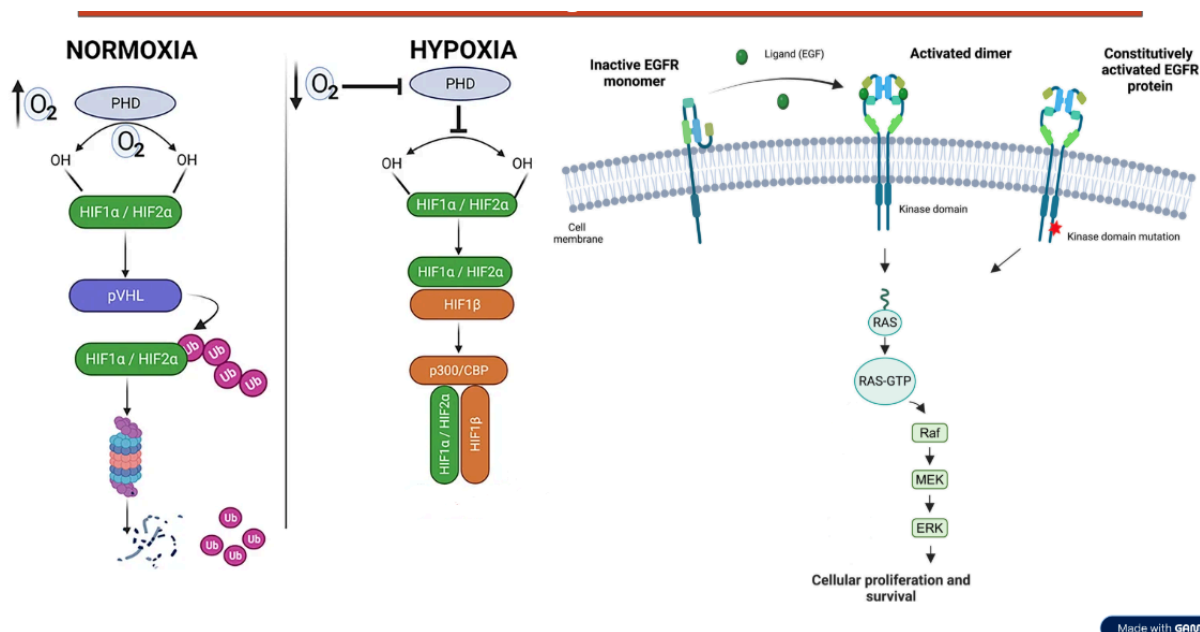
Traditionally, EGFR overexpression has been linked to gene amplification, where cancer cells produce extra copies of the EGFR gene, resulting in higher levels of the receptor. However, recent research has shown that EGFR can also increase in a different way. Under low oxygen conditions, known as hypoxia, proteins called Hypoxia-Inducible Factors (HIFs), especially HIF-2 α , become active. Instead of increasing the number of EGFR genes, HIF-2 α helps produce more EGFR protein by increasing the translation of existing EGFR mRNA through an alternative translation mechanism involving eIF4E2.

Although both gene amplification and hypoxia increase EGFR protein levels, they occur through different biological processes. It is still not well understood whether these two mechanisms produce the same signaling behavior or whether they respond differently to EGFR-targeted drugs. Understanding these differences is important because it may explain why some tumors become resistant to treatment even when they have high levels of EGFR.

In this project, a computational model of the EGFR–ERK signaling pathway was developed to compare hypoxia-driven translational EGFR upregulation with EGFR gene amplification. The model was used to

study how these two mechanisms affect ERK signaling and to evaluate their response to EGFR inhibitors. This work aims to improve our understanding of hypoxia-induced therapy resistance and provide insights that may support the development of more effective cancer treatment strategies.

BIOLOGICAL BACKGROUND



● Hypoxia and HIF Signaling

Hypoxia refers to a condition in which cells experience low oxygen levels. It is a common characteristic of solid tumors because rapidly growing cancer cells consume oxygen faster than new blood vessels can supply it. To survive under these conditions, cancer cells activate a group of proteins known as Hypoxia-Inducible Factors (HIFs), mainly HIF-1α and HIF-2α.

Under normal oxygen conditions (normoxia), HIF-1 α and HIF-2 α are continuously hydroxylated by Prolyl Hydroxylase Domain (PHD) enzymes. This hydroxylation allows the von Hippel–Lindau (pVHL) protein to recognize HIFs and mark them with ubiquitin molecules, leading to their degradation by the proteasome. As a result, HIF proteins remain at very low levels and hypoxia-responsive genes are not activated.

In contrast, under hypoxic conditions, the activity of PHD enzymes decreases because oxygen is required for their function. Without hydroxylation, HIF proteins escape degradation and accumulate inside the cell. HIF- α then binds with HIF-1 β and the transcriptional co-activator p300/CBP to form an active transcription complex. This complex enters the nucleus and activates the expression of genes involved in angiogenesis, metabolism, cell survival, and adaptation to hypoxic stress. Recent studies have also shown that HIF-2 α can enhance the translation of EGFR mRNA through the eIF4E2-dependent translation pathway, increasing EGFR protein levels without altering the EGFR gene copy number.

● **EGFR–ERK Signaling Pathway**

The Epidermal Growth Factor Receptor (EGFR) is a receptor tyrosine kinase located on the cell membrane. Under normal conditions, binding of Epidermal Growth Factor (EGF) to EGFR causes two receptor molecules to dimerize, leading to activation of the receptor's intracellular kinase domain through autophosphorylation.

Activated EGFR initiates several downstream signaling pathways, among which the RAS–RAF–MEK–ERK pathway is one of the most important. Activated EGFR recruits adaptor proteins that activate RAS by converting it into its GTP-bound active form (RAS-GTP). RAS-GTP

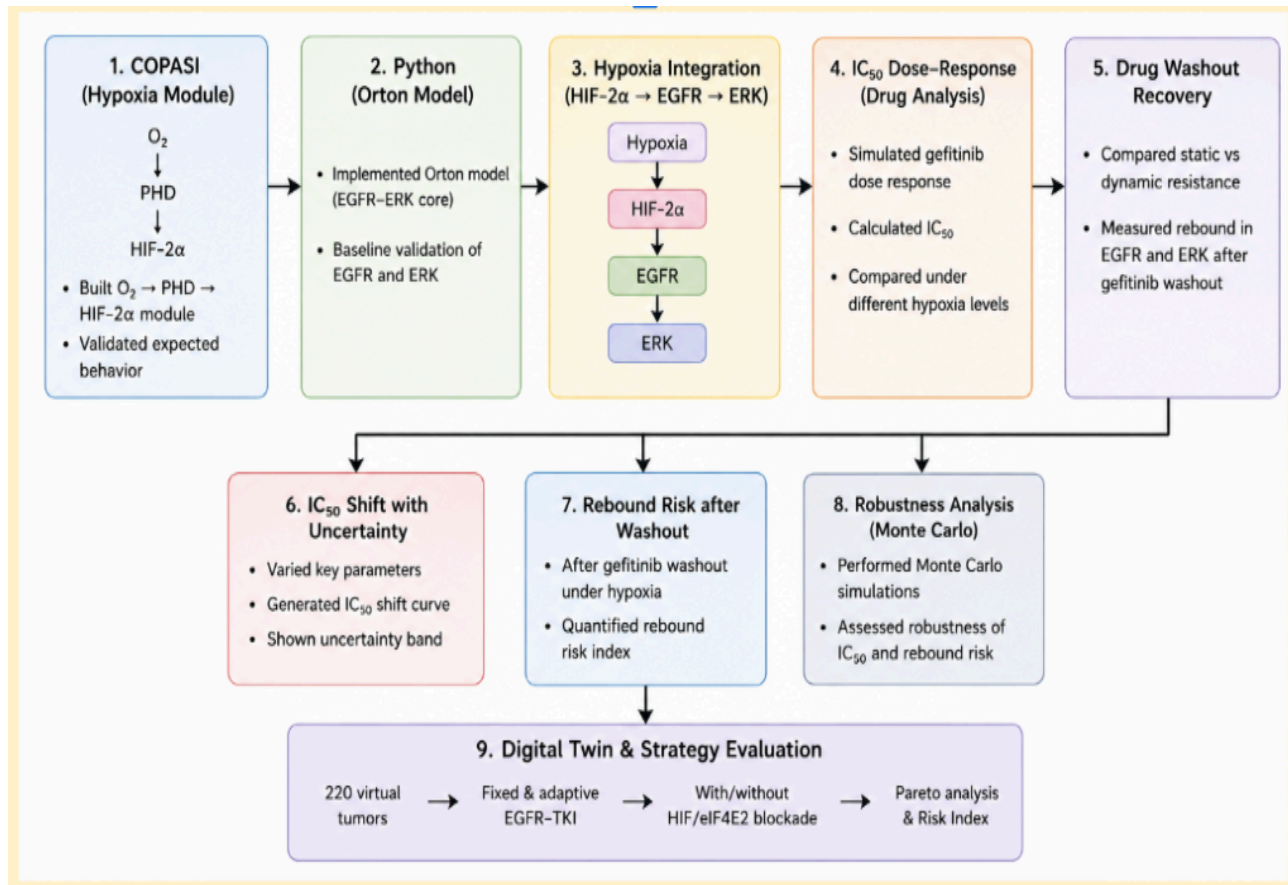
subsequently activates RAF kinase, which phosphorylates and activates MEK. MEK then activates ERK through phosphorylation.

Activated ERK translocates to the nucleus, where it regulates the expression of genes involved in cell proliferation, differentiation, and survival. Under normal physiological conditions, this signaling pathway is tightly regulated. However, in many cancers, EGFR becomes overexpressed or constitutively active due to gene amplification, activating mutations, or hypoxia-induced translational upregulation. Continuous activation of the EGFR–ERK pathway results in excessive cell proliferation, enhanced survival, and tumor progression, making EGFR an important target for cancer therapy.

OBJECTIVES

- Develop a mechanistic ODE model integrating the PHD HIF-2 α hypoxia axis with the canonical EGFR–ERK signaling pathway.
- Compare EGFR overexpression arising from hypoxia-driven translational upregulation versus gene amplification on:
 - ERK activation dynamics
 - Gefitinib IC₅₀
 - Drug washout rebound behaviour
- Evaluate whether targeting the hypoxia-translational axis (HIF-2 α /eIF4E2) improves therapeutic response compared with EGFR inhibition alone.

METHODOLOGY



The workflow of this study consists of nine major steps, starting from model development and ending with the evaluation of treatment strategies using a digital twin approach.

Step 1: Development of the Hypoxia Module (COPASI)

The first step was to build a hypoxia model in COPASI. This module represents how oxygen levels regulate the activity of HIF-2 α through Prolyl Hydroxylase Domain (PHD) enzymes. Under normal oxygen conditions, PHD promotes the degradation of HIF-2 α , whereas under hypoxia, reduced PHD activity allows HIF-2 α to accumulate. The module was validated to ensure that it behaved as expected under both normoxic and hypoxic conditions.

Step 2: Implementation of the EGFR–ERK Model (Python)

The core EGFR–ERK signaling pathway was implemented in Python using the reduced Orton model. This model includes the major signaling components responsible for transmitting signals from EGFR activation to ERK activation. Baseline simulations were performed to verify that the model reproduced normal EGFR and ERK signaling behavior.

Step 3: Integration of Hypoxia with EGFR Signaling

The validated hypoxia module was integrated with the EGFR–ERK signaling pathway. In this combined model, hypoxia increases HIF-2 α levels, which enhances EGFR protein production through translational regulation. The resulting increase in EGFR influences downstream ERK signaling, allowing the effects of hypoxia on pathway dynamics to be studied.

Step 4: Drug Dose–Response Analysis

Gefitinib, an EGFR inhibitor, was simulated at different concentrations to generate dose–response curves. From these simulations, the IC₅₀ value, which represents the drug concentration required to reduce pathway activity by 50%, was calculated. The analysis was repeated under different oxygen conditions to determine whether hypoxia alters drug sensitivity.

Step 5: Drug Washout Recovery

To study resistance after treatment, gefitinib was removed from the system after drug exposure. The recovery of EGFR and ERK activity was monitored over time. This helped compare how quickly signaling returned under normal oxygen conditions and under hypoxia, providing insight into dynamic drug resistance.

Step 6: IC₅₀ Uncertainty Analysis

Biological systems contain natural variability, and model parameters are not always known with complete certainty. Therefore, important parameters were varied within reasonable ranges to observe how the calculated IC₅₀ changed. This analysis produced an uncertainty band, indicating the reliability of the predicted drug response.

Step 7: Rebound Risk Analysis

Following drug washout, a rebound risk index was calculated to quantify how strongly EGFR and ERK signaling recovered. A higher rebound risk indicates a greater chance that tumor cells could rapidly regain proliferative signaling after treatment is stopped.

Step 8: Robustness Analysis

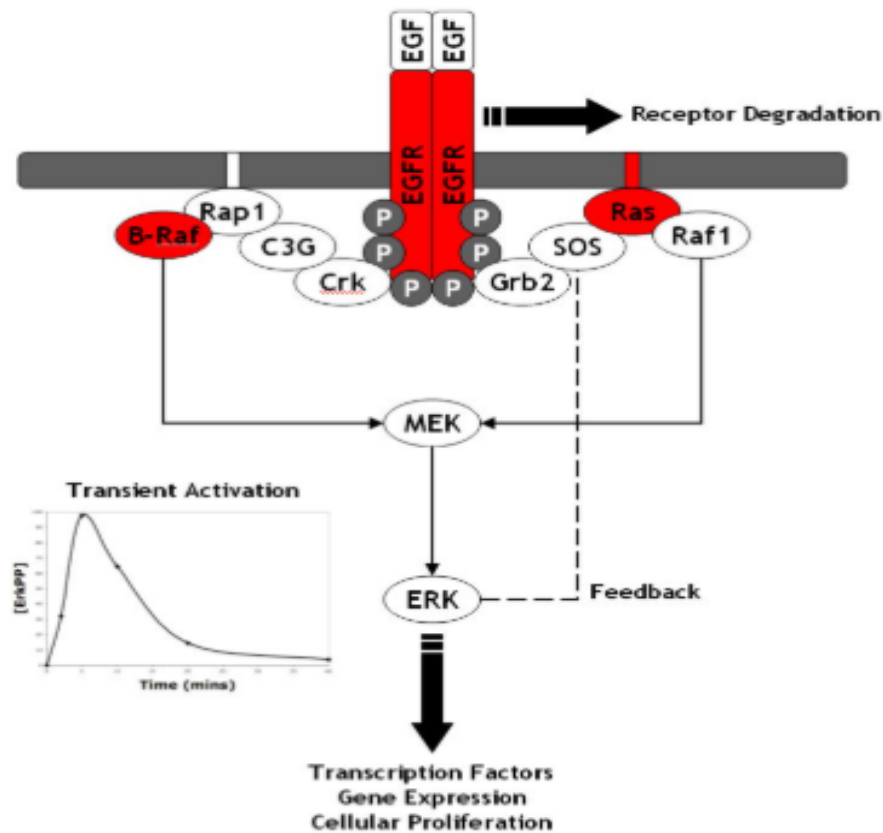
Monte Carlo simulations were performed by repeatedly varying model parameters to mimic biological variability among different tumors. This analysis tested whether the predicted IC₅₀ values and rebound risk remained consistent despite parameter uncertainty, thereby assessing the robustness of the model.

Step 9: Digital Twin and Treatment Strategy Evaluation

Finally, the complete model was used to create a virtual cohort of 220 tumors with different biological characteristics. Different treatment strategies, including fixed-dose and adaptive EGFR inhibitor therapy, were simulated with and without inhibition of the HIF/eIF4E2 translation pathway. The outcomes were compared using Pareto analysis and a factory-dependence (risk) index to identify the most effective strategies and distinguish hypoxia-driven resistance from resistance caused by EGFR gene amplification.

RESULTS

BASE MODEL: ORTON 2009



The core EGFR–ERK signaling pathway used in this study was adapted from the published Orton model, which was obtained from the BioModels database. This model represents the activation of EGFR by its ligand (EGF) and the subsequent signaling cascade through RAS, RAF, MEK, and ERK, ultimately leading to gene expression and cell proliferation. It also includes important regulatory mechanisms such as

receptor degradation and feedback regulation that control pathway activity.

The BioModels version served as the baseline model for this study because it is a well-established and experimentally validated representation of EGFR signaling. The model was further modified by integrating a hypoxia module (HIF-2 α -mediated EGFR regulation) to investigate the effect of hypoxia on ERK signaling dynamics and the response to EGFR-targeted therapy.

• DERIVED ODE EQUATIONS

$$\begin{aligned}
 (1) \quad PHD_{\text{act}} &= \frac{[O_2]}{K_{m,O_2} + [O_2]} && \text{PHD enzyme activity} \\
 (2) \quad \frac{d[HIF2\alpha]}{dt} &= k_{\text{syn,HIF}} - (k_{\text{deg,basal}} + k_{\text{deg,PHD}} \cdot PHD_{\text{act}}) \cdot [HIF2\alpha] && \text{HIF-2}\alpha \text{ Dynamics} \\
 (3) \quad f_{\text{HIF}} &= \frac{[HIF2\alpha]^{n_{\text{Hill}}}}{K_H^{n_{\text{Hill}}} + [HIF2\alpha]^{n_{\text{Hill}}}} && \text{Nonlinear activation threshold for egfr upregulation} \\
 (4) \quad \frac{d[EGFR]}{dt} &= A \cdot v_{\text{basal,syn}} + [\text{Other Orton Dynamics}] + k_{\text{syn,R}} \cdot f_{\text{HIF}} && \text{Hypoxia and gene amplification appended to orton 2009 EGFR equation}
 \end{aligned}$$

- Coupled into Orton 2009 via equation 4

To incorporate the effect of hypoxia into the EGFR–ERK signaling model, a simplified mathematical model describing HIF-2 α regulation was developed and coupled with the published Orton (2009) EGFR model.

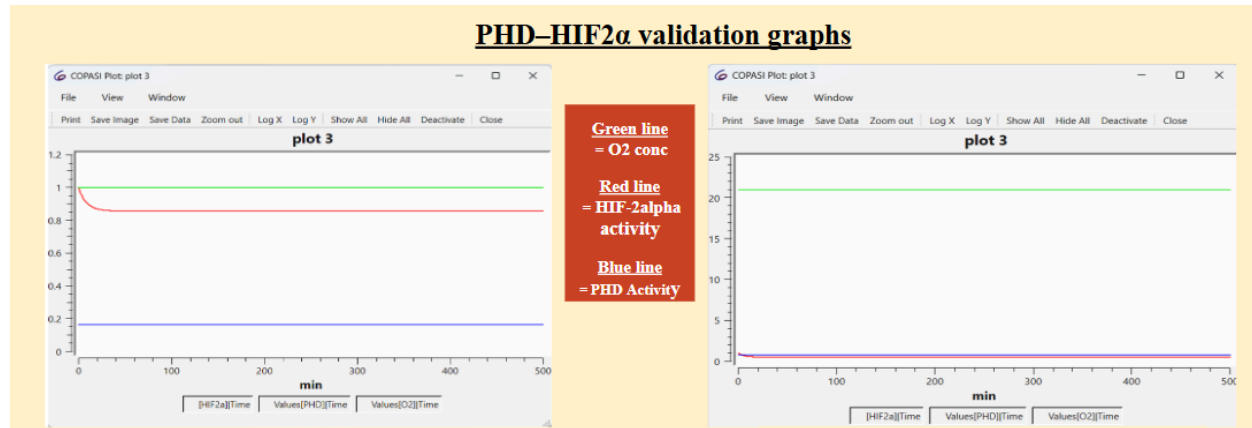
The first equation represents the activity of the Prolyl Hydroxylase Domain (PHD) enzyme as a function of oxygen concentration. Since PHD requires oxygen to function, its activity is high under normal oxygen conditions (normoxia) and decreases as oxygen levels fall during hypoxia.

The second equation describes the concentration of HIF-2 α over time. HIF-2 α is continuously synthesized within the cell but is degraded through two mechanisms: a constant basal degradation process and an oxygen-dependent degradation mediated by PHD. Under normoxic conditions, active PHD promotes rapid degradation of HIF-2 α , keeping its levels low. During hypoxia, reduced PHD activity decreases HIF-2 α degradation, allowing the protein to accumulate.

The third equation uses a Hill function to describe the effect of HIF-2 α on EGFR regulation. This function represents the nonlinear biological response, where small changes in HIF-2 α have little effect until a threshold is reached. Once sufficient HIF-2 α accumulates, EGFR expression increases rapidly.

The fourth equation integrates the hypoxia module with the original Orton (2009) EGFR model. In addition to the normal EGFR synthesis and degradation processes already present in the Orton model, an extra term was introduced to represent HIF-2 α -mediated translational upregulation of EGFR under hypoxic conditions. This modification enabled the model to simulate both normal EGFR signaling and hypoxia-induced EGFR overexpression within the same computational framework.

HYPOXIA MODULE(PHD-HIF-2-ALPHA)GRAPH

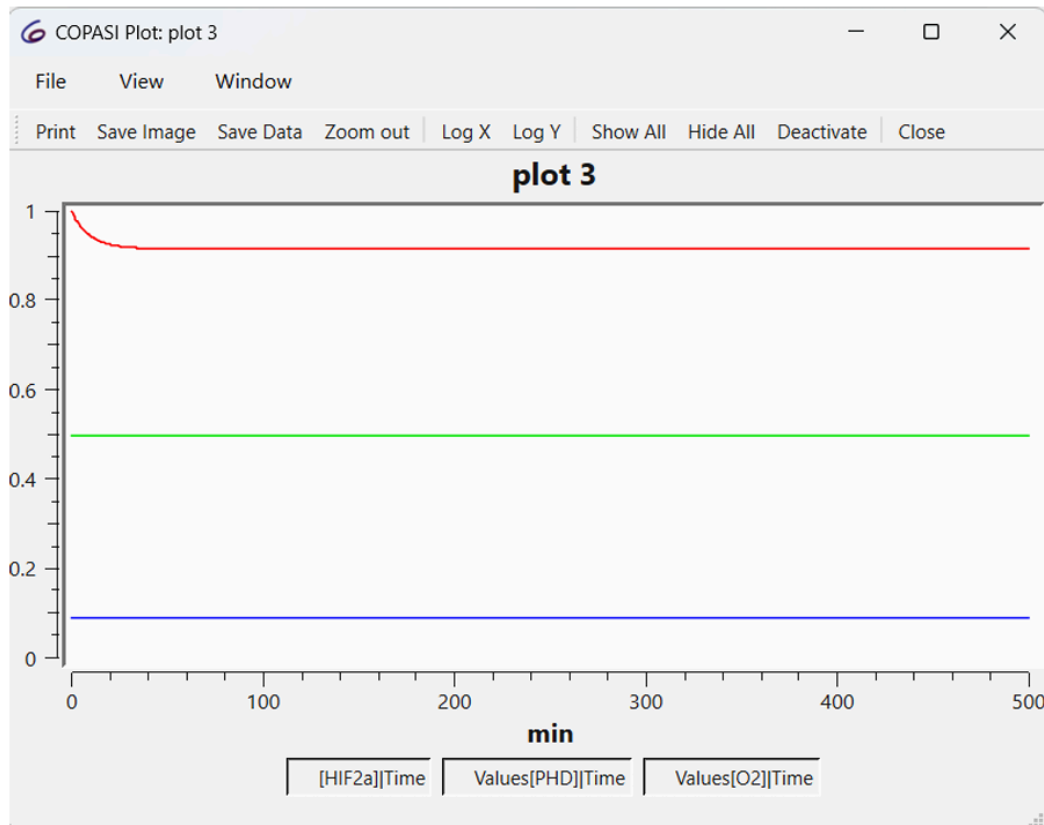


The hypoxia module was validated by simulating the relationship between oxygen concentration (O₂), PHD enzyme activity, and HIF-2 α levels under both hypoxic and normoxic conditions. The simulation results were consistent with the expected biological behavior.

In the hypoxic condition (left graph), oxygen concentration was maintained at a low level. As oxygen is required for PHD enzyme activity, the activity of PHD remained low throughout the simulation. Reduced PHD activity resulted in decreased degradation of HIF-2 α , allowing HIF-2 α levels to remain relatively high. This agrees with the biological mechanism in which hypoxia stabilizes HIF-2 α and enables it to regulate genes involved in cellular adaptation to low oxygen.

In the normoxic condition (right graph), oxygen concentration was maintained at a normal level. The availability of oxygen kept PHD enzyme activity high, leading to continuous degradation of HIF-2 α . As a result, HIF-2 α levels remained low during the simulation. This behavior is consistent with normal physiological conditions, where HIF-2 α does not accumulate because it is rapidly degraded.

Overall, the simulation results confirm that the developed hypoxia model accurately reproduces the expected oxygen-dependent regulation of HIF-2 α . This validated module was subsequently integrated with the EGFR–ERK signaling model to investigate the effect of hypoxia on EGFR expression and downstream signaling.



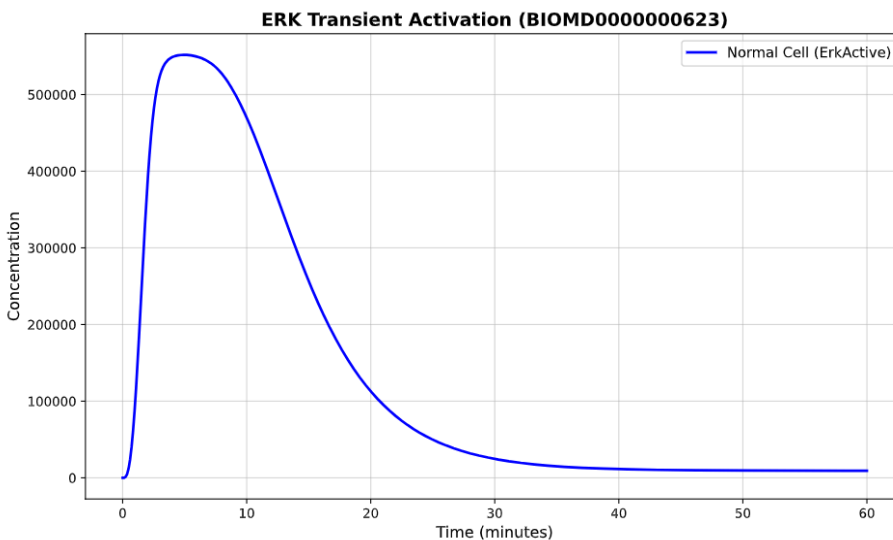
Above Figure shows the simulation results of the PHD–HIF-2 α module under hypoxic conditions. Oxygen concentration (green line) was maintained at a low and constant level throughout the simulation. Since PHD enzyme activity depends on the availability of oxygen, the reduced oxygen level resulted in low PHD activity (blue line).

As PHD is responsible for the oxygen-dependent degradation of HIF-2 α , its reduced activity allowed HIF-2 α to accumulate within the cell. This is

reflected by the red curve, where HIF-2 α rapidly reaches a steady-state concentration and remains stable over time. The slight decrease observed during the initial phase represents the system adjusting before reaching equilibrium.

The simulation follows the expected biological response to hypoxia, where low oxygen reduces PHD activity, preventing the degradation of HIF-2 α . The accumulated HIF-2 α can then regulate the expression and translation of hypoxia-responsive genes, including EGFR. These results confirm that the hypoxia module behaves as expected and is suitable for integration with the EGFR–ERK signaling model.

BASELINE VALIDATION – ORTON 2009 EGFR–ERK MODEL (BIOMD0000000623)



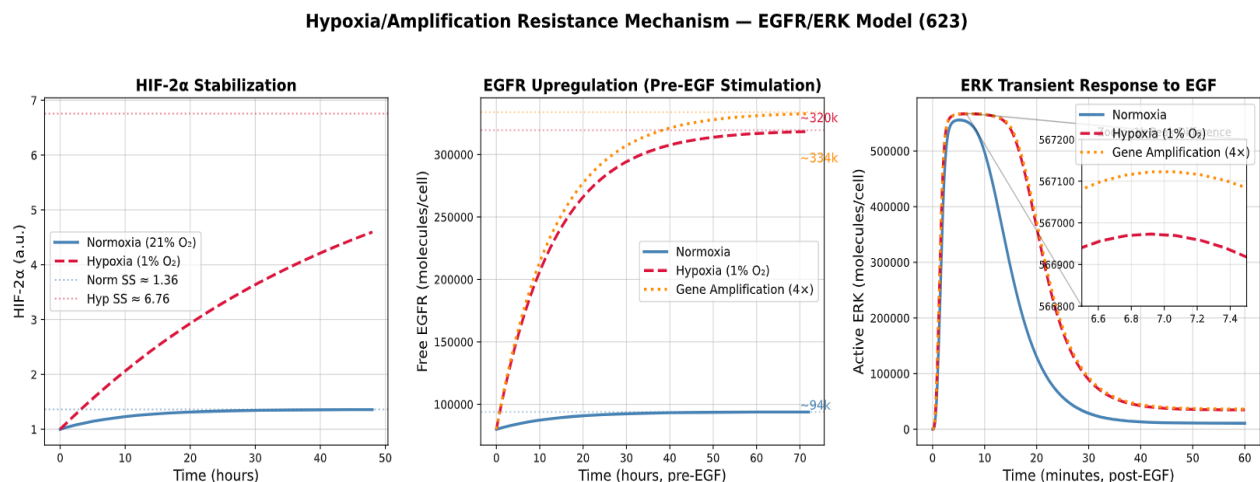
Before incorporating the hypoxia module, the original Orton 2009 EGFR–ERK signaling model (BIOMD0000000623) was validated to confirm that the core pathway dynamics were accurately reproduced. The model describes the EGFR–MAPK signaling cascade using

ordinary differential equations and includes the P90Rsk–Sos negative feedback loop responsible for transient ERK activation.

Simulation of the baseline model under normoxic conditions produced the characteristic transient ERK response reported in the original study. Following EGF stimulation, active ERK rapidly increased, reached a peak of approximately 5.5×10^5 molecules per cell, and subsequently declined toward basal levels within about 40 minutes. This behavior closely matched the published Orton 2009 results.

The successful reproduction of the canonical ERK transient demonstrates that the model implementation and feedback architecture are functioning correctly. This validation establishes a reliable foundation for introducing hypoxia-dependent HIF-2 α regulation and for conducting subsequent analyses of EGFR upregulation, drug response, and resistance mechanisms.

HYPOXIA VALIDATION (HYPOXIA $\rightarrow$ HIF-2 α $\rightarrow$ EGFR $\rightarrow$ ERK)



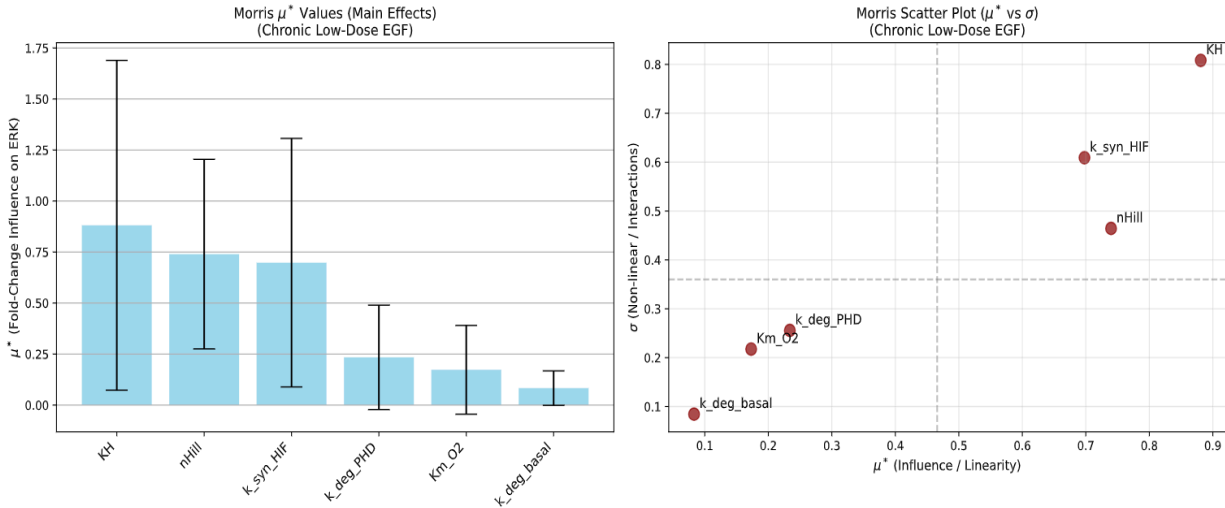
To validate the proposed hypoxia extension, the complete signaling chain linking oxygen deprivation to ERK activation was simulated.

Under hypoxic conditions (1% O₂), reduced PHD activity stabilized HIF-2 α , resulting in a progressive increase in HIF-2 α protein levels over 48 hours. In contrast, normoxic conditions maintained low HIF-2 α levels due to continuous oxygen-dependent degradation. This behavior confirmed that the oxygen-sensing module correctly captures the biological response to hypoxia.

The stabilized HIF-2 α promoted EGFR synthesis through the hypoxia-regulated translational pathway, leading to a substantial increase in receptor abundance over 72 hours. Hypoxic cells reached EGFR levels comparable to those produced by the simulated 4 $\times$ gene amplification scenario, whereas normoxic cells maintained considerably lower receptor levels. These results demonstrate that hypoxia can generate significant EGFR overexpression through a non-mutational mechanism.

The increase in receptor abundance amplified downstream signaling, producing a stronger and more sustained ERK activation response following EGF stimulation. Since a larger number of receptors were available for ligand binding, hypoxic and gene-amplified cells exhibited enhanced ERK activation compared with normoxic cells. Together, these findings validate the mechanistic pathway of Hypoxia $\rightarrow$ HIF-2 α Stabilization $\rightarrow$ EGFR Upregulation $\rightarrow$ Enhanced ERK Signaling implemented in the model.

MORRIS GLOBAL SENSITIVITY ANALYSIS



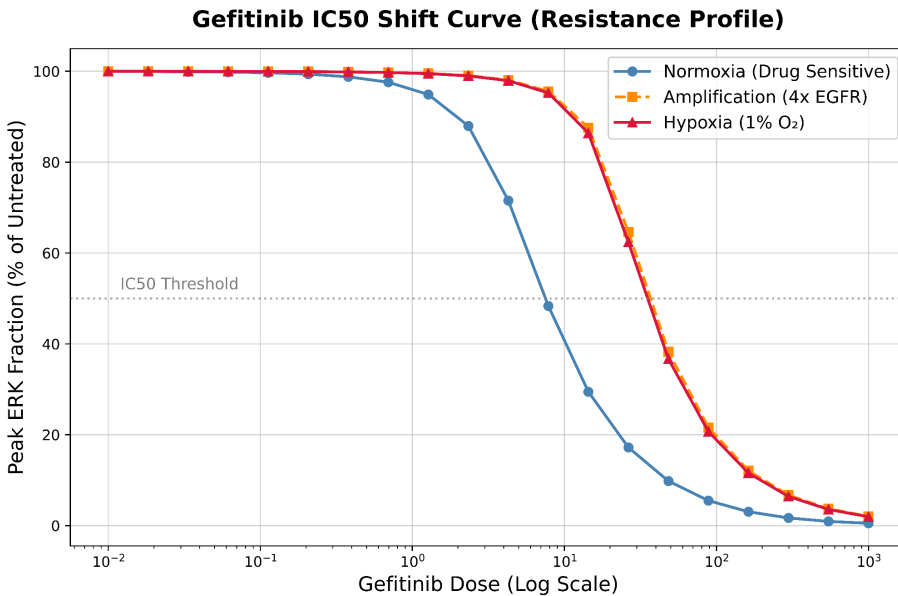
A Morris global sensitivity analysis was performed to identify the parameters that most strongly influence hypoxia-induced ERK amplification. Approximately 100 trajectories were sampled across six hypoxia-module parameters, and the model output was defined as the 48-hour ERK AUC fold change under hypoxic conditions relative to normoxia during chronic low-dose EGF stimulation. The Morris method quantifies parameter importance using μ^* (mean absolute elementary effect) and σ (standard deviation of elementary effects), representing overall influence and interaction/nonlinearity, respectively.

The analysis revealed that the parameters **K_H** and **k_{syn}HIF** exert the strongest influence on hypoxia-driven signaling amplification. Both parameters exhibited high μ^* values, indicating that they are major determinants of ERK activation under hypoxic conditions. In contrast, oxygen-sensing parameters such as **K_{O2}**, **k_{deg}PHD**, and **k_{deg}basal** showed relatively small effects on the model output.

The σ – μ^* relationship further indicated that **K_H** participates in significant nonlinear effects and parameter interactions, whereas **k_{syn}HIF** primarily acts as a direct driver of system behavior. Overall, the results suggest that HIF-2 α production and its downstream activation

threshold are the dominant regulators of hypoxia-induced EGFR/ERK resistance, highlighting these processes as critical control points within the model.

IC₅₀ DOSE-RESPONSE ANALYSIS (QUANTIFYING DRUG RESISTANCE)



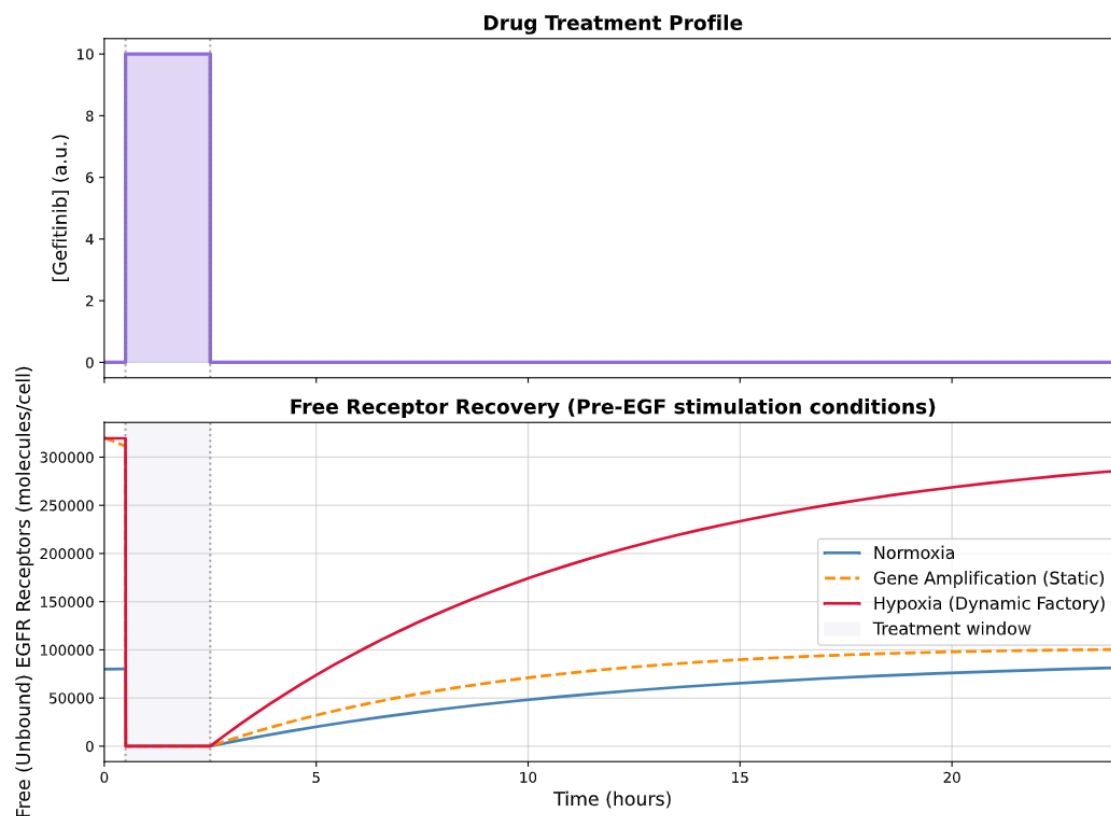
To quantify the impact of hypoxia-induced EGFR upregulation on therapeutic response, a dose-response analysis was performed using Gefitinib across a range of inhibitor concentrations. The sustained ERK activity remaining after treatment was used as a measure of pathway inhibition, and dose-response curves were generated for normoxic, hypoxic, and EGFR-amplified conditions.

The results showed a clear rightward shift in the dose-response curves for both hypoxic and gene-amplified cells relative to the normoxic baseline, indicating reduced drug sensitivity. Despite arising through different biological mechanisms, hypoxia-driven EGFR upregulation and 4× EGFR gene amplification produced nearly identical resistance

profiles, with comparable increases in the effective drug concentration required to suppress ERK signaling.

These findings suggest that receptor abundance is the primary determinant of resistance during continuous EGFR-targeted therapy, regardless of whether the increased receptor levels originate from dynamic hypoxia-mediated translation or static gene amplification. Therefore, hypoxic tumors may exhibit clinically significant Gefitinib resistance even in the absence of genetic alterations in the EGFR locus.

DRUG WASHOUT RECOVERY EXPERIMENT (STATIC VS DYNAMIC RESISTANCE)



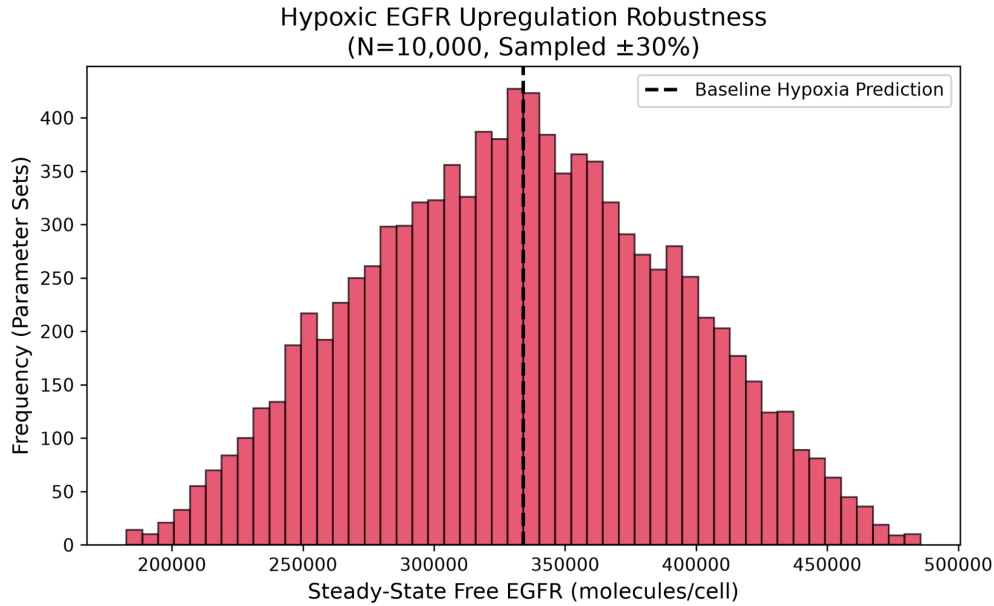
To investigate the temporal behavior of resistance mechanisms, a drug washout experiment was performed following Gefitinib treatment. The simulation consisted of three phases: a baseline period, a treatment phase

during which EGFR signaling was suppressed, and a washout phase in which the drug was removed and pathway recovery was monitored. The objective was to compare the recovery dynamics of hypoxia-driven EGFR upregulation and EGFR gene amplification.

During treatment, both hypoxic and gene-amplified conditions exhibited similar suppression of ERK signaling, indicating comparable sensitivity under continuous drug exposure. However, substantial differences emerged after drug removal. Hypoxic cells rapidly restored EGFR receptor levels through ongoing HIF-2 α /eIF4E2-mediated synthesis, leading to a faster recovery of downstream signaling. In contrast, gene-amplified cells recovered more gradually because they lacked an active receptor re-synthesis mechanism.

These results demonstrate that hypoxia generates a dynamic and reversible form of resistance that extends beyond simple receptor overexpression. While hypoxia and gene amplification appear similar during treatment, their post-treatment behavior differs significantly. The rapid rebound observed under hypoxic conditions suggests that drug-free intervals may strongly influence therapeutic outcomes and highlights the importance of considering treatment timing in addition to drug dosage.

MONTE CARLO ROBUSTNESS ANALYSIS



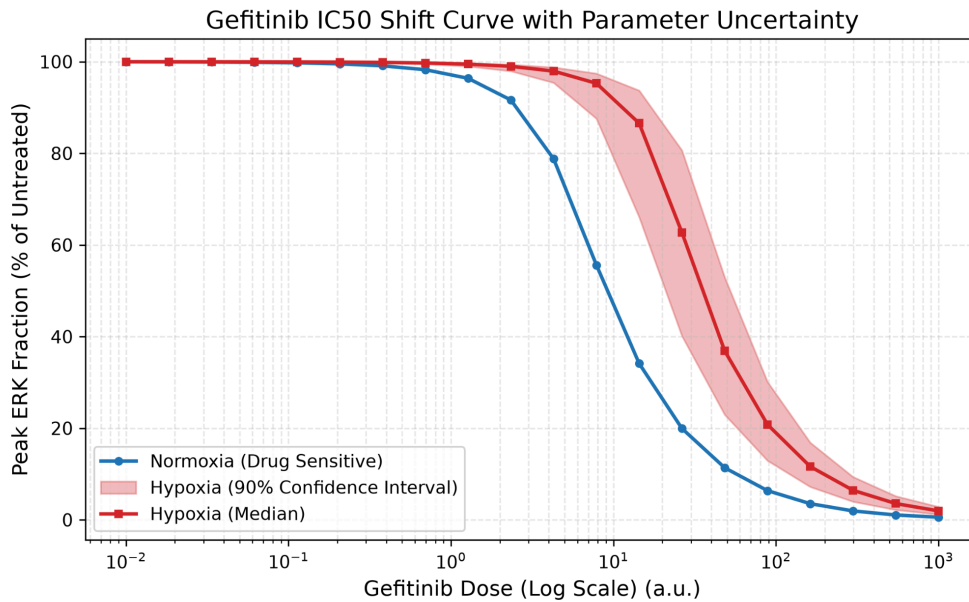
To evaluate the robustness of the hypoxia-induced resistance phenotype, a Monte Carlo analysis was performed by varying the key hypoxia-module parameters K_H , n_{Hill} , and $k_{syn,HIF}$ across biologically plausible ranges. Multiple parameter combinations were sampled to assess whether the observed increase in EGFR expression and ERK activation remained consistent under parameter uncertainty.

The resulting distribution of model outputs showed that the resistance phenotype was preserved across the majority of sampled parameter sets. Although variations in parameter values produced quantitative differences in the magnitude of the response, the overall trend of hypoxia-driven EGFR upregulation and enhanced downstream signaling remained unchanged.

These results demonstrate that the model predictions are not dependent on a single parameter set and are robust to moderate uncertainty in the most influential hypoxia-related parameters. The persistence of the resistance phenotype across the sampled parameter space increases

confidence in the biological conclusions drawn from the model and supports the reliability of subsequent analyses.

IC₅₀ SHIFT CURVE WITH PARAMETER UNCERTAINTY



To determine whether the observed drug-resistance phenotype remains stable under parameter variability, the IC₅₀ dose–response analysis was repeated across multiple parameter combinations generated through Monte Carlo sampling. The influential hypoxia-module parameters were varied within biologically plausible ranges, and the resulting dose–response curves were compared to evaluate the robustness of the predicted resistance shift.

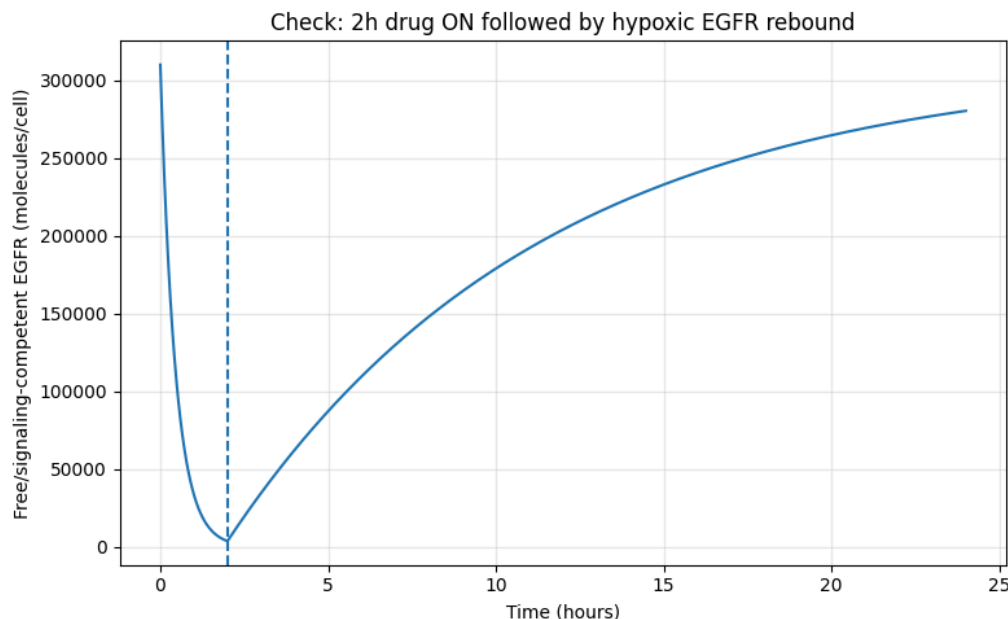
Across the sampled parameter sets, hypoxic conditions consistently exhibited a rightward shift in the dose–response curve relative to normoxia, indicating reduced sensitivity to Gefitinib. Although the magnitude of the shift varied between simulations, the overall resistance phenotype remained preserved throughout the parameter space. This demonstrates that hypoxia-driven EGFR upregulation reliably increases

the drug concentration required to achieve equivalent pathway inhibition.

The results indicate that the predicted resistance behavior is robust to parameter uncertainty and is not dependent on a narrow set of model assumptions. The persistence of the IC_{50} shift across multiple simulations strengthens confidence in the conclusion that hypoxia-induced translational upregulation can generate clinically relevant EGFR inhibitor resistance comparable to that produced by gene amplification.

QUANTIFYING HYPOXIC EGFR REBOUND RISK AFTER GEFITINIB WASHOUT

1. EGFR RECOVERY DYNAMICS FOLLOWING GEFITINIB WASHOUT



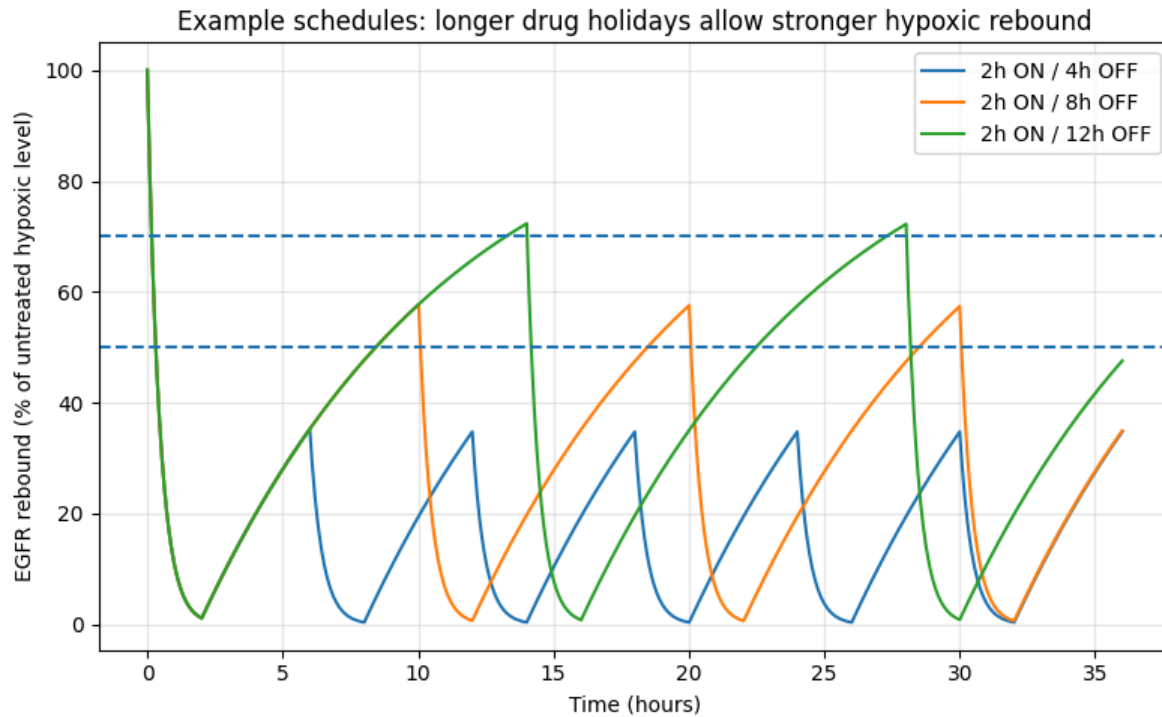
Following Gefitinib treatment, the recovery dynamics of hypoxic EGFR were investigated to quantify receptor rebound after drug removal. At

the start of the simulation, hypoxic cells maintained elevated EGFR levels of approximately 310,000 molecules per cell. During the 2-hour treatment period, active EGFR signaling was strongly suppressed, causing signaling-competent receptor levels to decrease toward zero.

Once the drug was removed, EGFR levels gradually recovered toward the hypoxic steady-state value due to ongoing HIF-2 α /eIF4E2-mediated receptor synthesis. The recovery profile demonstrated that hypoxic cells retain the ability to continuously regenerate receptor pools even after successful pathway inhibition.

These results indicate that EGFR suppression achieved during treatment may be temporary if sufficient recovery time is allowed. The observed rebound behavior highlights the importance of considering post-treatment receptor dynamics when designing therapeutic schedules for hypoxic tumors.

2. EFFECT OF DRUG HOLIDAYS ON HYPOXIC EGFR REBOUND

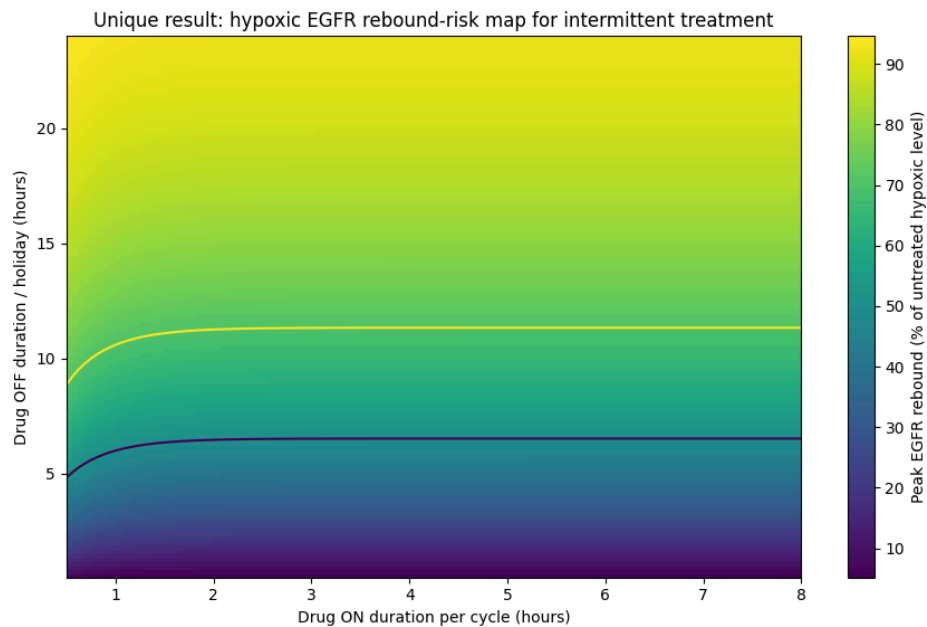


To evaluate the impact of treatment timing, the relationship between drug-free intervals and receptor recovery was examined. The simulations showed that hypoxia-driven resistance is influenced not only by drug dosage but also by the duration of OFF periods following treatment.

After drug withdrawal, hypoxic cells behaved as an active receptor-production system, continuously rebuilding EGFR pools through hypoxia-dependent translation. As OFF-period duration increased, receptor recovery became progressively stronger, increasing the likelihood of signaling reactivation and therapeutic escape.

These findings suggest that prolonged drug holidays may unintentionally promote resistance by allowing hypoxic tumors sufficient time to restore receptor abundance. Therefore, treatment scheduling should consider both drug exposure and recovery intervals to effectively control rebound signaling.

3. REBOUND-RISK MAP FOR TREATMENT SCHEDULING



A rebound-risk map was generated to identify combinations of drug ON and OFF durations that limit EGFR recovery after treatment. The analysis quantified the maximum OFF periods that could be tolerated while keeping receptor rebound below predefined thresholds.

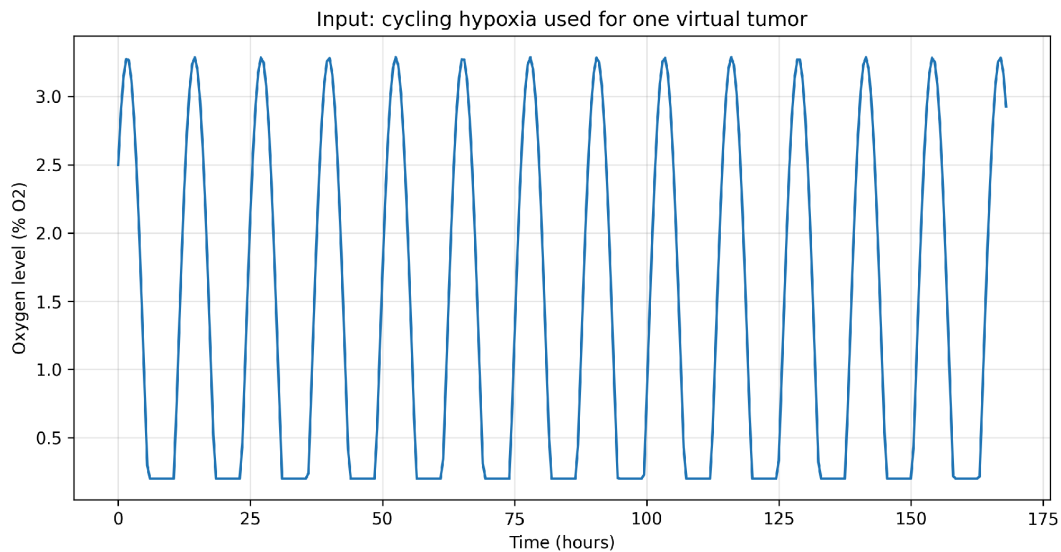
The results revealed that increasing treatment duration beyond approximately two hours provided only marginal improvements in rebound control. In contrast, OFF-period duration had a much larger effect on receptor recovery. For example, maintaining rebound below 50% required OFF periods of approximately 6 hours or less, whereas longer drug holidays produced substantially greater recovery.

Overall, the rebound-risk map demonstrates that OFF-period duration is the dominant determinant of hypoxic receptor rebound once sufficient pathway suppression has been achieved. These findings provide

practical guidance for optimizing treatment schedules aimed at minimizing resistance re-emergence.

MECHANISM-AWARE DIGITAL TWIN OF HYPOXIA-DRIVEN EGFR RESISTANCE: DISTINGUISHING DYNAMIC TRANSLATION-DEPENDENT ESCAPE FROM STATIC GENE AMPLIFICATION

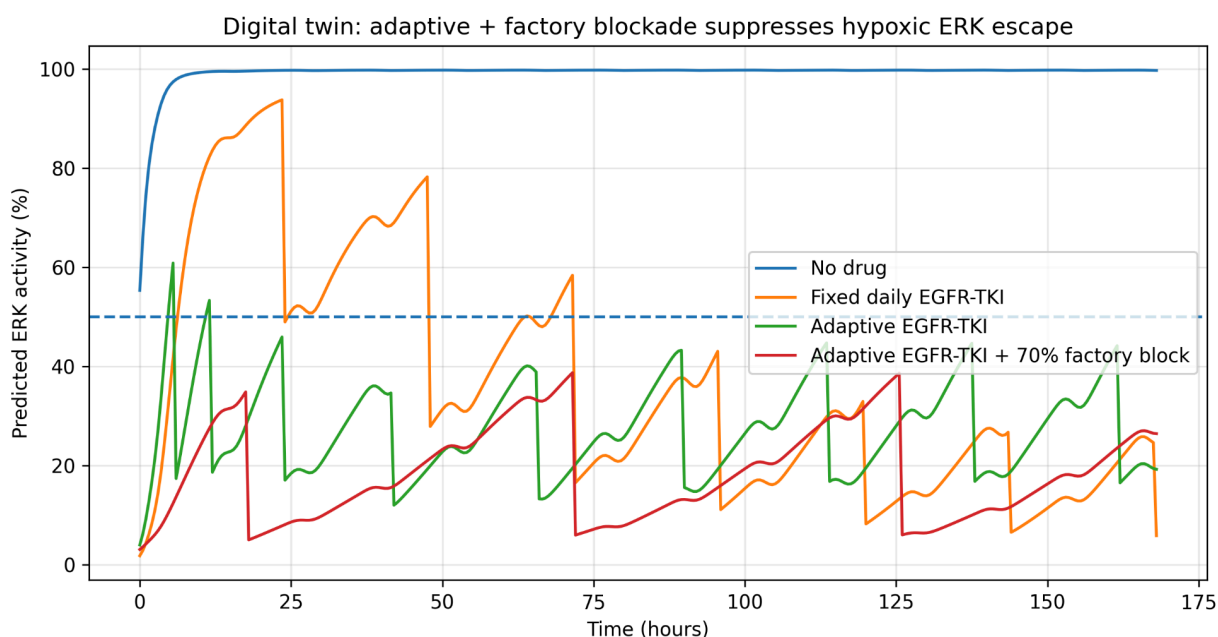
1. CYCLING HYPOXIA INPUT



The digital twin framework was driven using a dynamic oxygen profile designed to mimic the fluctuating oxygen conditions commonly observed within solid tumors. Rather than assuming constant hypoxia, oxygen levels were allowed to vary continuously over the seven-day simulation period, producing alternating phases of mild and severe hypoxia. This approach better reflects the unstable vascular supply and heterogeneous microenvironment found in real tumors.

These oxygen fluctuations directly influenced HIF-2 α stabilization and subsequent EGFR production within the model. As oxygen availability changed over time, the signaling network continuously adapted, creating a more realistic representation of hypoxia-driven resistance than static oxygen conditions. This dynamic input formed the foundation for all subsequent digital twin simulations and treatment-response analyses.

2. SINGLE-TUMOR ERK STRATEGY COMPARISON

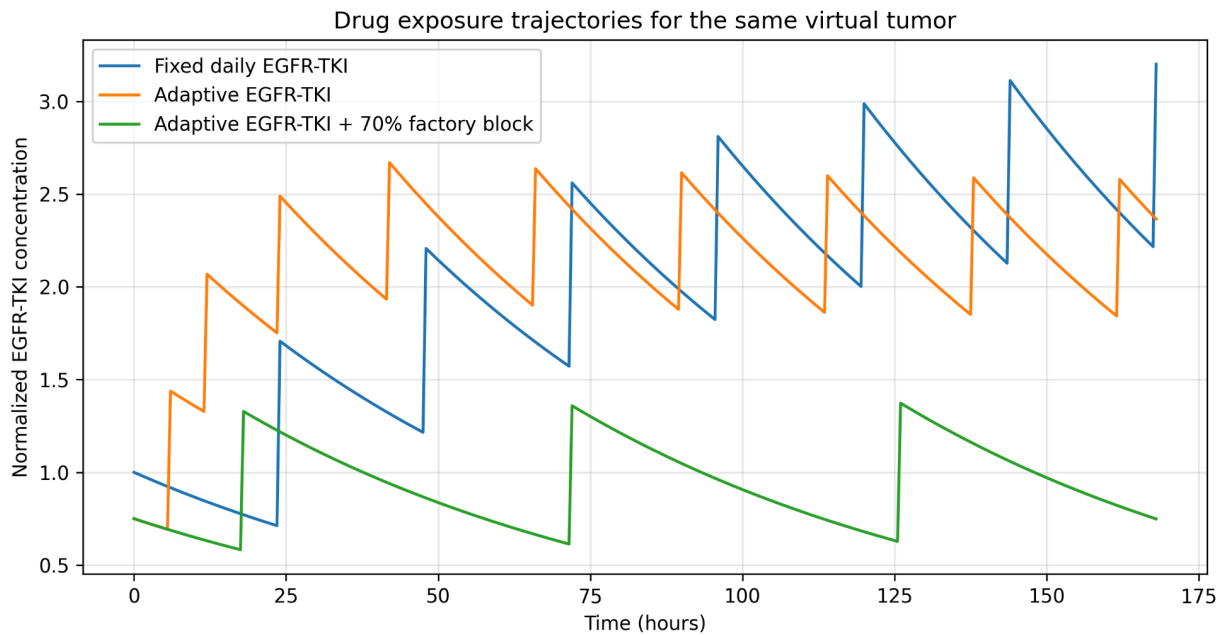


To evaluate the effectiveness of different therapeutic approaches, ERK signaling dynamics were compared in a representative virtual tumor under four treatment strategies: no treatment, fixed daily EGFR-TKI therapy, adaptive EGFR-TKI therapy, and adaptive EGFR-TKI combined with translation-factory blockade. The untreated tumor maintained persistent ERK activity, whereas treatment strategies progressively improved signaling control.

Among the tested approaches, adaptive EGFR-TKI combined with factory blockade produced the strongest suppression of ERK activity. By

targeting both the signaling pathway and the hypoxia-driven receptor production mechanism, this strategy minimized ERK escape events and maintained pathway control more effectively than conventional fixed-dose treatment. These findings suggest that mechanism-aware therapies may provide superior control of hypoxia-driven resistance.

3. DRUG EXPOSURE TRAJECTORIES

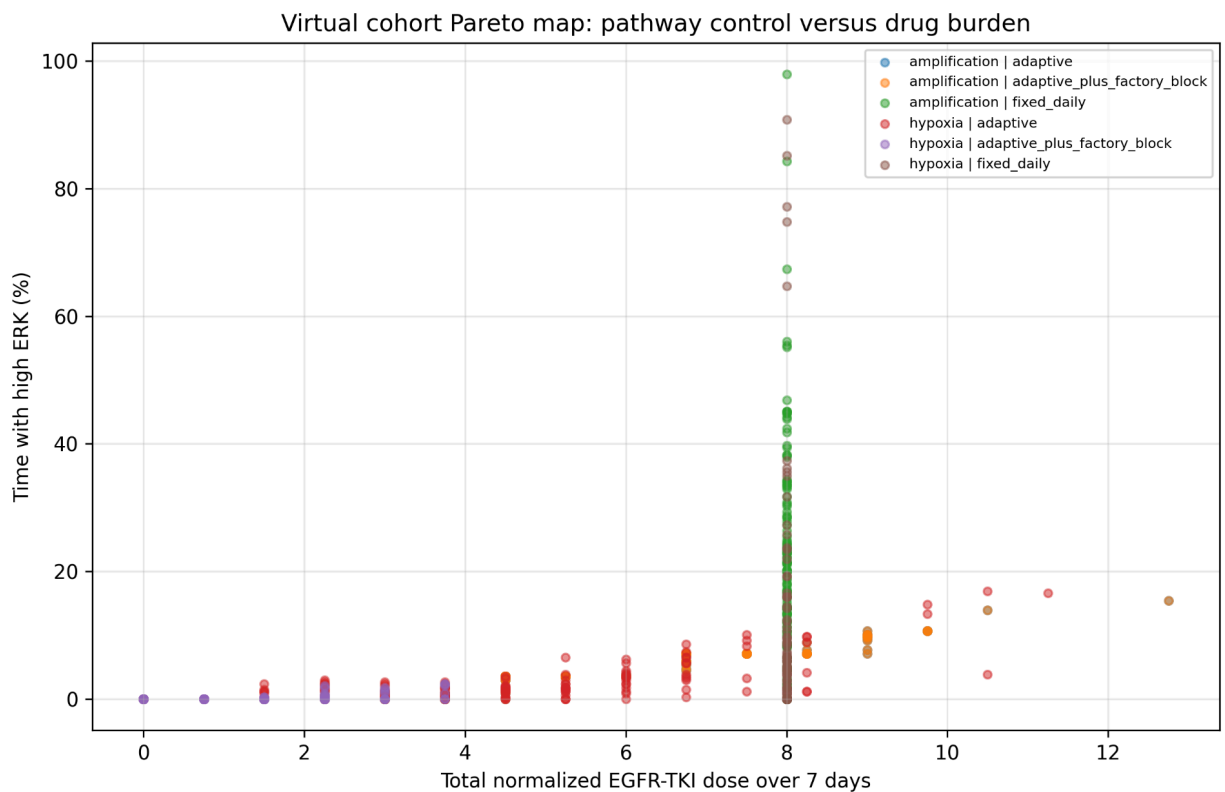


Drug exposure trajectories were analyzed to determine the amount of therapy required under each treatment strategy over the seven-day simulation period. Fixed daily treatment produced continuous drug administration regardless of tumor state, whereas adaptive strategies adjusted drug delivery in response to ERK activity. This allowed a direct comparison between therapeutic effectiveness and total drug usage.

The results demonstrated that adaptive treatment schedules could maintain effective signaling control while reducing unnecessary drug exposure. By administering therapy only when required, adaptive approaches achieved a more efficient balance between efficacy and

treatment burden. This highlights the potential of feedback-controlled treatment strategies to improve outcomes while limiting excessive drug use.

4. VIRTUAL COHORT PARETO MAP

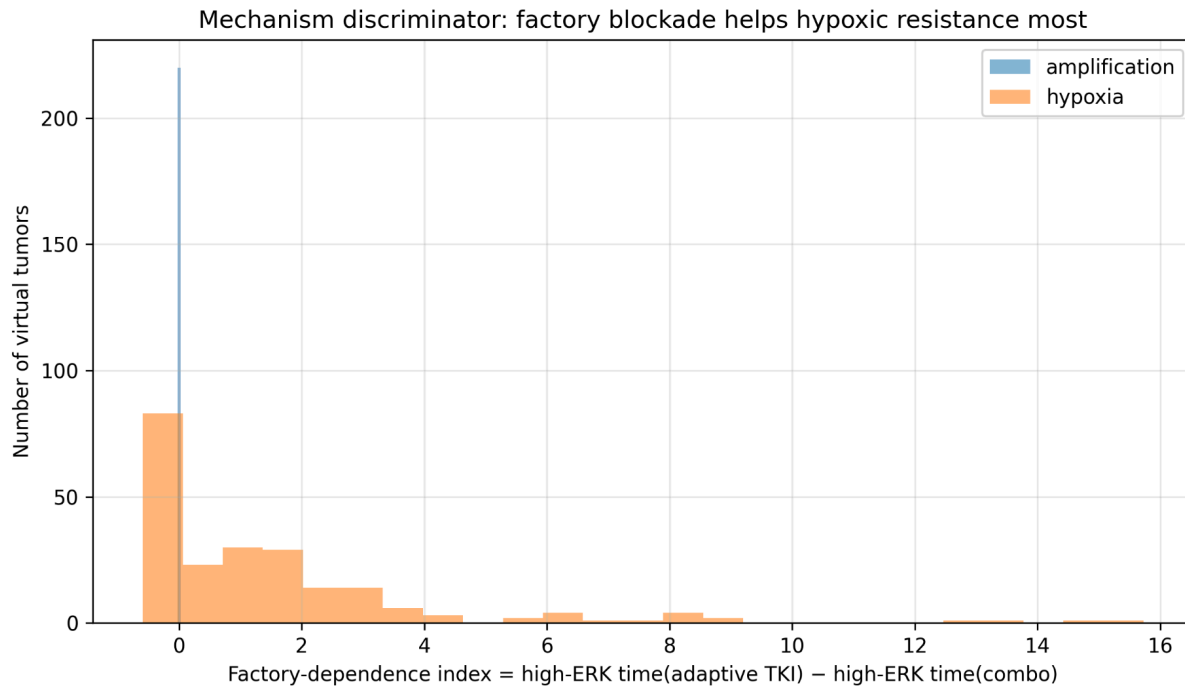


A virtual cohort consisting of 220 tumors was generated to evaluate treatment performance across a heterogeneous population. Each point on the Pareto map represented an individual tumor, with total drug exposure plotted on the x-axis and the percentage of time ERK remained highly active plotted on the y-axis. This visualization enabled simultaneous assessment of therapeutic efficacy and treatment intensity.

The Pareto analysis revealed a clear trade-off between drug usage and signaling suppression. Tumors positioned closer to the lower-left region of the plot achieved both low ERK activity and reduced drug exposure,

representing optimal therapeutic outcomes. The map therefore provides a useful framework for identifying treatment strategies that maximize efficacy while minimizing treatment burden across diverse tumor populations.

5. FACTORY DEPENDENCE CLASSIFIER



To distinguish tumors driven by hypoxia-mediated receptor production from those driven by static amplification, a Factory Dependence Index was developed. This metric quantified the reduction in high-ERK time achieved when translation-factory blockade was added to adaptive EGFR-TKI therapy. Higher values indicated stronger dependence on the HIF-2 α /eIF4E2-mediated EGFR synthesis pathway.

The classifier successfully separated tumors based on their underlying resistance mechanism. Tumors exhibiting high factory dependence derived substantial benefit from combined therapy, whereas tumors with low dependence showed little additional improvement beyond EGFR

inhibition alone. These results support the central hypothesis that the biological route leading to EGFR overexpression can influence therapeutic response and can be identified using mechanism-aware digital twin analysis.

CONCLUSION

This study demonstrates that not all EGFR overexpression is biologically equivalent. Although hypoxia-driven translational upregulation and gene amplification produce similar receptor levels and comparable resistance to Gefitinib during continuous treatment, the underlying mechanisms create fundamentally different cellular behaviors over time. While gene amplification represents a relatively static increase in receptor abundance, hypoxia generates a dynamic and reversible resistance state sustained by ongoing HIF-2 α /eIF4E2-mediated EGFR synthesis.

The simulations show that these differences become most apparent after drug withdrawal. Hypoxic cells rapidly rebuild EGFR pools and restore ERK signaling, whereas gene-amplified cells recover more slowly due to the absence of an active synthesis program. As a result, treatment schedules that appear equally effective during drug exposure can lead to markedly different outcomes during drug-free intervals. The rebound-risk analysis further suggests that limiting OFF periods may be more important than extending treatment duration in hypoxic tumors.

Most importantly, the digital twin framework predicts that combining EGFR-targeted therapy with HIF/eIF4E2 pathway inhibition can selectively eliminate ERK escape in hypoxic tumors while providing little benefit in gene-amplified tumors. This highlights a key principle emerging from the study: the mechanism responsible for EGFR overexpression may be just as important as the level of EGFR itself when selecting therapeutic strategies.

Although these findings are based on computational modeling and require experimental validation, they suggest that hypoxia-driven resistance represents a unique and potentially targetable vulnerability.

Overall, this work provides a mechanistic framework for distinguishing different routes to EGFR overexpression and demonstrates how digital twin approaches can support more personalized and mechanism-informed cancer therapy.

LIMITATIONS

- All findings are computational predictions within a reduced ODE model and have not been validated by wet-lab experiments.
- The digital twin uses simplified receptor pool and ERK activation equations rather than the full Orton signaling network.
- The 70% HIF/eIF4E2 factory blockade is a virtual intervention with no specific clinical drug assigned to it.
- Parameter values for the hypoxia module are literature-derived and have not been directly fitted to experimental data from any specific cell line.

FUTURE DIRECTIONS

- Couple the hypoxia–EGFR–ERK digital twin to a minimal EMT circuit incorporating core nodes such as Snail, ZEB1, and E-cadherin.
- Introduce spatial compartmentalization by separating the tumor into at least a hypoxic core and an oxygenated rim to capture microenvironmental heterogeneity.
- Validate co-expression of hypoxic EGFR and HIF-2 α signatures using publicly available scRNA-seq or spatial transcriptomics datasets.

- Rebuild the full hypoxia extension within the complete Orton/COPASI backbone and conduct comprehensive parameter sensitivity and bifurcation analysis.

INDIVIDUAL CONTRIBUTIONS

Team Member	Contribution
Pragata Ghosh	Conducted biological background research, reviewed literature on hypoxia-driven EGFR regulation, and formulated the project hypothesis, research question, objectives, and biological rationale.
Pranjal Salil Shekhar	Performed mathematical modelling validation of the PHD–HIF-2 α oxygen-sensing module under normoxic, hypoxic, and severe hypoxic conditions using COPASI to integrate hypoxic modules into Orton model, derived ODE equations using COPASI. Designed the overall computational methodology pipeline .
Meenu S A	Developed the Orton 2009 EGFR–ERK base model framework, derived and integrated the four hypoxia-extension ODE equations, and conducted the Monte Carlo robustness analysis and IC ₅₀ uncertainty quantification studies.
Adbhutha Beere	Performed baseline ERK model validation, validated the hypoxia signaling cascade (HIF-2 α stabilization → EGFR upregulation → ERK activation), conducted the Morris global sensitivity analysis, and performed the IC ₅₀ dose–response and drug washout recovery experiments to characterize resistance dynamics.
Harsh Upadhyay	Developed the mechanism-aware digital twin framework, quantified hypoxic EGFR rebound

	risk following Gefitinib washout, generated and analyzed the virtual cohort of 220 tumors, performed treatment strategy comparisons, developed the Pareto map and Factory Dependence Classifier, and formulated the final conclusions and therapeutic insights.
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